Deploying Microservices on AWS ECS Fargate

Étape 1: Create a Task Definition

1. **Navigate to the ECS Console:**
    - Open the AWS Management Console and go to the **ECS** service.
2. **Create a Task Definition:**
    - Choose **Task Definitions** from the left menu.
    - Click on **Create new Task Definition** and select **Fargate**.
3. **Configure Task Definition:**
    - **Task Name:** Provide a meaningful name for the task.
    - **Container Definitions:** Add details about your container(s):
        - **Container Name:** Enter a name for your container.
        - **Image URI:** Provide the Docker image URI (e.g., from ECR or Docker Hub).
        - **Port Mappings:** Map the container port (e.g., `80`) to the host port.
    - Set the desired **CPU and Memory** requirements.
    - Add any required **environment variables**, **volumes**, or **logging configurations**.
4. **Save Task Definition:**
    - Click **Create** to save the task definition.

Étape 2: Create a Cluster

1. **Go to Clusters:**
    - In the ECS console, select **Clusters** from the left menu.
2. **Create a New Cluster:**
    - Click on **Create Cluster** and choose **Networking only (Fargate)**.
    - Provide a cluster name and configure the networking options if necessary.
3. **Save Cluster:**
    - Click **Create** to finalize the cluster setup.

Étape 3: Create a Target Group

1. **Navigate to the EC2 Console:**
    - Open the AWS EC2 service and go to the **Target Groups** section under **Load Balancing**.
2. **Create a New Target Group:**
    - Click **Create target group**.
    - Choose **Application Load Balancer** as the target type.
    - Under the **Target type**, select **IP addresses**.
    - Provide the necessary details, such as:
        - **Target Group Name**
        - **Protocol:** Typically HTTP or HTTPS.
        - **Port:** The port number your application listens on (e.g., `80`).
3. **Configure Health Checks:**
    - Set the health check path (e.g., `/health` or `/`).
4. **Create Target Group:**
    - Click **Create** to save.

Étape 4: Create an Application Load Balancer

1. **Go to Load Balancers in EC2 Console:**
    - Under **Load Balancing**, click **Load Balancers**.
2. **Create a New Load Balancer:**
    - Click **Create Load Balancer** and select **Application Load Balancer**.
    - Specify the following:
        - **Name**: Enter a name for the load balancer.
        - **Scheme**: Choose Internet-facing or Internal, depending on the use case.
        - **Listeners**: Add HTTP (or HTTPS if SSL is configured).
        - **VPC and Subnets**: Select the VPC and subnets for the load balancer.
3. **Configure Target Group:**
    - Associate the load balancer with the target group created in Étape 3.
4. **Create Load Balancer:**
    - Review the configuration and click **Create**.

Étape 5: Create an ECS Service

1. **Return to ECS Console:**
    - In the ECS service, go to **Clusters** and select the cluster you created.
2. **Create a Service:**
    - Click **Create Service** and select:
        - **Launch Type:** Fargate.
        - **Task Definition:** Choose the task definition created in Étape 1.
        - **Service Name:** Provide a meaningful name.
        - **Number of Tasks:** Specify the desired count.
3. **Configure Load Balancer:**
    - Under **Load Balancing**, select **Application Load Balancer**.
    - Link the service to the target group created in Étape 3.
4. **Networking:**
    - Select the appropriate VPC and subnets.
    - Attach necessary security groups that allow inbound traffic on the required ports.
5. **Autoscaling (Optional):**
    - Configure autoscaling policies if needed to handle variable traffic.
6. **Create Service:**
    - Review the configuration and click **Create Service**.

Exposing Application with a Load Balancer

Retrieve the Load Balancer DNS Name:

- Navigate to the **EC2 Console** in the AWS Management Console.
- Under the **Load Balancing** section, click on **Load Balancers**.
- Locate your Application Load Balancer in the list and copy its **DNS name**.